



Model 623 Survey Cart Assembly Guide

Mobilizing your system on the GSSI Model 623 survey cart allows you to cover ground much more easily and conveniently than with the conventional system configuration. The 623 cart incorporates a survey wheel for high-precision distance measurements. The cart can accommodate the 400 MHz, 900 MHz, 1.0 GHz, 1.5/1.6 GHz and the 2.6 GHz antennas. For the 1.0, 1.6, and 2.6 GHz antenna, you will need a survey wheel adaptor cable. It is GSSI part number FG62X/5100B CBL.

Contact your GSSI sales representative for more details. GSSI can be reached at (603) 893-1109 Monday-Friday, 8:30 AM – 5 PM US Eastern Time.

Tools Required: None.

Unfold Cart and Attach the Wheels

- 1** Unfold the cart frame and insert the black tips of the top assembly into the receivers on the front wheel fork.



- 2** Squeeze the wheel-lock clamps, and insert the wheel shaft into the axle. It will slide all the way in and lock securely. To remove wheels, squeeze the wheel-lock clamp again and pull the wheel straight out. Note: The wheels are foam filled and do not require inflation.



- 3** Slide the front wheel onto the front wheel fork and tighten clamp. When the clamp is slightly tight, turn the handle to the locked position. This will further tighten the clamp. Exercise care not to over-tighten the front wheel as this may result in damage to the front fork.



- 4** Ensure that the survey wheel encoder is correctly positioned on the inside of rear left wheel and that the survey wheel is making contact with the tire's rim.

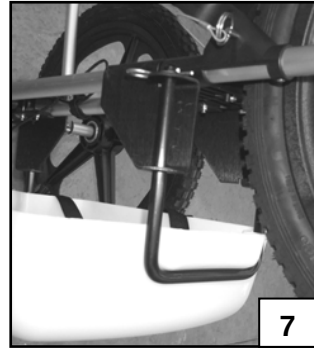


Mount the Antenna

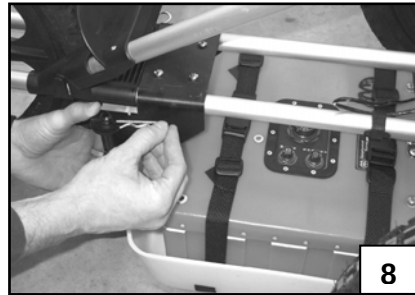
- 5** Place the antenna into the white plastic tub and secure with the attached straps. If you are using the cart system with an antenna of higher frequency than 900 MHz, place the antenna in the bottom of the tub and stick the white plastic plate with the Velcro strips to the Velcro on top of the antenna. Secure the assembly with straps. If you are using the 623 cart with the 400 MHz or 900 MHz antenna, the white Velcro plate is not needed.



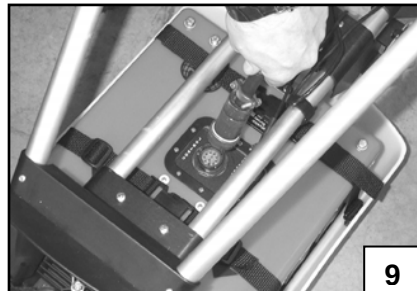
- 6** With the arrows on the top of the antenna housing pointing toward the front of the cart, place the tub under the cart so that the tub handles face the front and the back of the cart.
- 7** Lift the tub to fit the white fiberglass brackets under the handle and insert them through the 2 double holes on the frame.



- 8** Secure with the metal pins. The antenna tub should just touch the ground surface. It is intended to be loose because it needs to be able to float over small obstacles. If you are using a small, higher frequency antenna, be sure that the antenna is centered in the tub.



- 9** Connect the female end of the control cable to port that is labeled CONTROL, connect the lead from the survey wheel (4 pin) to the SURVEY port. These leads should only be hand tightened.



Attach the SIR-3000

- 10** Hold the SIR-3000 in the mounting bracket so that all of the holes line up. Beginning with the top 2 holes, attach the SIR-3000 with the thumb screws. Pivot the SIR-3000 back and forth to get the most comfortable viewing position, and screw in the bottom 2 thumb screws. Be sure to only hand-tighten the screws. If you purchased the optional sunshade, be sure that it is positioned on the outside of the bracket rather than being sandwiched between the mounting bracket and the SIR-3000.



- 11** Attach the male end of the control cable to the antenna port on the back of the SIR-3000. Be sure to only hand-tighten this connection as over tightening may cause system damage.



Note: Your survey wheel is only as accurate as your calibration, so be sure to calibrate your wheel often. Survey wheel calibration instructions can be found in your SIR-Systems manual.